

Supplementary Table 5. Permutation importance by feature family and top individual predictors

Frontiers in Signal Processing R1 supplementary table

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Standalone display version for LaTeX upload/review. Machine-readable source: `source_tables/Supplementary_Table_5_prediction_explainability.csv`. Rows: 48; columns: 16. The source table is split below into source-defined sections and column blocks so that all non-empty cells remain readable.

table block: feature family importance

Table 1: Supplementary Table 5, feature family importance, column block 1 of
2

model	feature family	positive importance sum AUPRC	positive importance mean AUPRC	positive importance sum AUROC
Classical signal features	clinical signal	0.1752	0.0146	0.1751
Signal + SSL	clinical signal	0.4874	0.0406	0.2016
Signal + SSL	SSL embedding	2.6	0.0406	0.7787
Signal + SSL + phenotype	clinical signal	0.3688	0.0307	0.1921
Signal + SSL + phenotype	phenotype	0.0235	0.0235	0.0159
Signal + SSL + phenotype	SSL embedding	1.893	0.0296	0.7695

Table 2: Supplementary Table 5, feature family importance, column block 2 of 2

model	feature family	positive importance mean AUROC	n features
Classical signal features	clinical signal	0.0146	12
Signal + SSL	clinical signal	0.0168	12
Signal + SSL	SSL embedding	0.0122	64
Signal + SSL + phenotype	clinical signal	0.016	12
Signal + SSL + phenotype	phenotype	0.0159	1
Signal + SSL + phenotype	SSL embedding	0.012	64

table block: top feature importance

Table 3: Supplementary Table 5, top feature importance, column block 1 of 3

model	feature family	feature	importance mean AUPRC	importance std AUPRC
Classical signal features	clinical signal	FHR mean	0.0579	0.0329
Classical signal features	clinical signal	FHR missing fraction	0.0466	0.0258
Classical signal features	clinical signal	deceleration proxy count	0.0328	0.0491
Classical signal features	clinical signal	FHR p05	0.019	0.0199
Classical signal features	clinical signal	FHR UC corr	0.0189	0.0232
Classical signal features	clinical signal	UC missing fraction	0	0
Classical signal features	clinical signal	UC sd	-9.85e-06	0.0002
Classical signal features	clinical signal	FHR sd	-0.0007	0.0029
Classical signal features	clinical signal	UC mean	-0.0017	0.0078
Classical signal features	clinical signal	FHR p95	-0.0019	0.0024
Classical signal features	clinical signal	UC p95	-0.0061	0.0078
Classical signal features	clinical signal	FHR p50	-0.0275	0.031
Signal + SSL	clinical signal	FHR missing fraction	0.1642	0.0343
Signal + SSL	SSL embedding	emb 55	0.1584	0.0346
Signal + SSL	SSL embedding	emb 32	0.1387	0.0273
Signal + SSL	SSL embedding	emb 17	0.1179	0.0318
Signal + SSL	SSL embedding	emb 19	0.1178	0.0266
Signal + SSL	SSL embedding	emb 43	0.1004	0.0317
Signal + SSL	clinical signal	FHR mean	0.0989	0.0314
Signal + SSL	SSL embedding	emb 18	0.0886	0.0271
Signal + SSL	SSL embedding	emb 63	0.0818	0.0321
Signal + SSL	SSL embedding	emb 52	0.0783	0.0193
Signal + SSL	SSL embedding	emb 26	0.0775	0.0203
Signal + SSL	SSL embedding	emb 25	0.0769	0.0223
Signal + SSL	SSL embedding	emb 42	0.0709	0.0247
Signal + SSL	SSL embedding	emb 44	0.0699	0.0323
Signal + SSL	SSL embedding	emb 29	0.068	0.0248
Signal + SSL + phenotype	clinical signal	FHR missing fraction	0.1508	0.0354
Signal + SSL + phenotype	SSL embedding	emb 55	0.1404	0.037
Signal + SSL + phenotype	SSL embedding	emb 32	0.1263	0.0276
Signal + SSL + phenotype	SSL embedding	emb 17	0.1102	0.0307
Signal + SSL + phenotype	SSL embedding	emb 19	0.0895	0.0259
Signal + SSL + phenotype	SSL embedding	emb 43	0.0803	0.0308
Signal + SSL + phenotype	clinical signal	FHR mean	0.0801	0.0304
Signal + SSL + phenotype	SSL embedding	emb 63	0.0683	0.0323
Signal + SSL + phenotype	SSL embedding	emb 52	0.0673	0.0227
Signal + SSL + phenotype	SSL embedding	emb 26	0.0627	0.0206
Signal + SSL + phenotype	SSL embedding	emb 25	0.062	0.0234
Signal + SSL + phenotype	SSL embedding	emb 18	0.0613	0.0291
Signal + SSL + phenotype	SSL embedding	emb 42	0.0554	0.0234
Signal + SSL + phenotype	SSL embedding	emb 53	0.0534	0.0223

Signal + SSL + phenotype	SSL embedding	emb 44	0.0527	0.0304
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Table 4: Supplementary Table 5, top feature importance, column block 2 of 3

model	feature family	importance mean AUROC	importance std AUROC	feature label
Classical signal features	clinical signal	0.0736	0.0204	Mean FHR
Classical signal features	clinical signal	0.0344	0.0288	FHR missing fraction
Classical signal features	clinical signal	0.0508	0.0365	Deceleration-proxy burden
Classical signal features	clinical signal	0.0144	0.0138	FHR 5th percentile
Classical signal features	clinical signal	-0.0044	0.0182	FHR-UC correlation
Classical signal features	clinical signal	0	0	UC missing fraction
Classical signal features	clinical signal	0.0001	0.0002	UC variability (SD)
Classical signal features	clinical signal	0.0017	0.0015	FHR variability (SD)
Classical signal features	clinical signal	-0.0002	0.0058	Mean uterine contraction
Classical signal features	clinical signal	2.97e-05	0.0003	FHR 95th percentile
Classical signal features	clinical signal	-0.0011	0.0059	UC 95th percentile
Classical signal features	clinical signal	-0.0103	0.0168	Median FHR
Signal + SSL	clinical signal	0.0846	0.0327	FHR missing fraction
Signal + SSL	SSL embedding	0.0702	0.0246	SSL dimension 55
Signal + SSL	SSL embedding	0.0503	0.0224	SSL dimension 32
Signal + SSL	SSL embedding	0.0187	0.0239	SSL dimension 17
Signal + SSL	SSL embedding	0.0119	0.0316	SSL dimension 19
Signal + SSL	SSL embedding	0.0335	0.0221	SSL dimension 43
Signal + SSL	clinical signal	0.0508	0.0162	Mean FHR
Signal + SSL	SSL embedding	0.025	0.0174	SSL dimension 18
Signal + SSL	SSL embedding	0.0358	0.0306	SSL dimension 63
Signal + SSL	SSL embedding	0.0112	0.0185	SSL dimension 52
Signal + SSL	SSL embedding	0.0128	0.0194	SSL dimension 26
Signal + SSL	SSL embedding	0.0647	0.0237	SSL dimension 25
Signal + SSL	SSL embedding	-0.0051	0.0204	SSL dimension 42
Signal + SSL	SSL embedding	-0.0278	0.021	SSL dimension 44
Signal + SSL	SSL embedding	0.014	0.0151	SSL dimension 29
Signal + SSL + phenotype	clinical signal	0.0868	0.0327	FHR missing fraction
Signal + SSL + phenotype	SSL embedding	0.0721	0.0241	SSL dimension 55
Signal + SSL + phenotype	SSL embedding	0.0523	0.0224	SSL dimension 32
Signal + SSL + phenotype	SSL embedding	0.0214	0.0233	SSL dimension 17
Signal + SSL + phenotype	SSL embedding	0.009	0.0295	SSL dimension 19
Signal + SSL + phenotype	SSL embedding	0.0332	0.0214	SSL dimension 43
Signal + SSL + phenotype	clinical signal	0.05	0.0156	Mean FHR
Signal + SSL + phenotype	SSL embedding	0.0372	0.031	SSL dimension 63
Signal + SSL + phenotype	SSL embedding	0.0137	0.0189	SSL dimension 52
Signal + SSL + phenotype	SSL embedding	0.0117	0.0188	SSL dimension 26
Signal + SSL + phenotype	SSL embedding	0.0638	0.0243	SSL dimension 25
Signal + SSL + phenotype	SSL embedding	0.023	0.0156	SSL dimension 18
Signal + SSL + phenotype	SSL embedding	-0.0049	0.0197	SSL dimension 42
Signal + SSL + phenotype	SSL embedding	0.0538	0.0192	SSL dimension 53

Signal + SSL + phenotype	SSL embedding	-0.0268	0.0202	SSL dimension 44
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Table 5: Supplementary Table 5, top feature importance, column block 3 of 3

model	feature family	rank AUPRC	rank AUROC
Classical signal features	clinical signal	1	1
Classical signal features	clinical signal	2	3
Classical signal features	clinical signal	3	2
Classical signal features	clinical signal	4	4
Classical signal features	clinical signal	5	11
Classical signal features	clinical signal	6	8
Classical signal features	clinical signal	7	6
Classical signal features	clinical signal	8	5
Classical signal features	clinical signal	9	9
Classical signal features	clinical signal	10	7
Classical signal features	clinical signal	11	10
Classical signal features	clinical signal	12	12
Signal + SSL	clinical signal	1	1
Signal + SSL	SSL embedding	2	2
Signal + SSL	SSL embedding	3	6
Signal + SSL	SSL embedding	4	18
Signal + SSL	SSL embedding	5	31
Signal + SSL	SSL embedding	6	10
Signal + SSL	clinical signal	7	5
Signal + SSL	SSL embedding	8	13
Signal + SSL	SSL embedding	9	8
Signal + SSL	SSL embedding	10	32
Signal + SSL	SSL embedding	11	29
Signal + SSL	SSL embedding	12	3
Signal + SSL	SSL embedding	13	58
Signal + SSL	SSL embedding	14	76
Signal + SSL	SSL embedding	15	26
Signal + SSL + phenotype	clinical signal	1	1
Signal + SSL + phenotype	SSL embedding	2	2
Signal + SSL + phenotype	SSL embedding	3	5
Signal + SSL + phenotype	SSL embedding	4	17
Signal + SSL + phenotype	SSL embedding	5	35
Signal + SSL + phenotype	SSL embedding	6	9
Signal + SSL + phenotype	clinical signal	7	6
Signal + SSL + phenotype	SSL embedding	8	8
Signal + SSL + phenotype	SSL embedding	9	23
Signal + SSL + phenotype	SSL embedding	10	29
Signal + SSL + phenotype	SSL embedding	11	3
Signal + SSL + phenotype	SSL embedding	12	15
Signal + SSL + phenotype	SSL embedding	13	58
Signal + SSL + phenotype	SSL embedding	14	4

